Mangrove Notes: Decoupling Format and Content

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Abstract: Current Personal Knowledge Management Systems (PKMS) often trap knowledge within rigid formatting paradigms and isolated file-hierarchy silos, creating friction for collaborative information sharing. To address these structural limitations, we introduce Mangrove, an exploratory note-taking system that decouples format from content to improve individual sensemaking and team collaboration. Rather than utilizing static pages or files, Mangrove stores all information as discrete nodes within a graph database tree. Nodes are presented in dynamic "projections" that render the exact same underlying nodes in multiple visual formats, such as prose, nested outlines, or tables. Mangrove nodes can also possess multiple parents, allowing a single piece of information to simultaneously show across disparate contexts without duplication. The system also uses customizable key-value properties and schemas to bridge the gap between freeform notes and structured, queryable data. Ultimately, this flexible graph-based approach eliminates friction when exporting knowledge and improves human recall and AI-driven discovery across collaborative infrastructures.

1 Introduction

Through modern technology, digital note-taking has evolved to help millions of users seamlessly capture information. These use cases span a wide spectrum: structured learning (class notes, flashcards, wikis), active sensemaking (brainstorming, research, outlining, drafting), and data logging (task tracking, journaling, notes on media/people). To support such varied workflows, many methodologies have emerged, driving the development of highly specialized note-taking applications and formats. Using these tools, an individual gradually constructs a Personal Knowledge Management System (PKMS)—often termed a "second brain" [1]. This system not only facilitates the quick and serendipitous recall of past recorded information, but also deepens comprehension as the user distills thoughts into text.

Yet, the inherent “personal” nature of a PKMS often turns it into an isolated silo, creating severe friction when users attempt to share their knowledge. This disconnect stems from both cognitive habits and technical constraints. Cognitively, years of individualized use result in highly idiosyncratic note-taking structures that are intuitive to the creator but intractable to outside collaborators. Technically, many applications lock content into rigid formatting paradigms and offer limited export controls (e.g., formatting and context may be lost when pasting into an email). Even tools like Obsidian [2], which promote interoperability through local file-based Markdown, simultaneously impose hidden structural boundaries via the hierarchical directory-file storage model. Moreover, information captured in one format (e.g., a nested outline) cannot fluidly be recontextualized into another (e.g., flowing prose). These structural limitations trap knowledge at the individual level, making it more difficult to share insights in important collaborative environments [3].

This friction is particularly detrimental today; as artificial intelligence increasingly automates team communication and search tasks, a robust PKMS remains vital. It not

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only serves as the personalized ground truth required to steer AI tools away from generic output, but also preserves the irreplaceable cognitive process of human sensemaking that cannot be outsourced without degrading the user's own understanding [4].

This paper introduces *Mangrove*, an exploratory note-taking system that decouples format from content, increasing flexibility in and sharability of a user's PKMS and allowing them to focus primarily on substance instead of syntax. Mangrove does away with the notion of pages or files, instead storing all information in a graph database that forms a tree of *nodes*, where each node represents one logical unit (e.g., a 'block', such as one paragraph or bullet point). This architecture is being actively developed and supports three key features:

(1) **Projections**. Nodes do not possess an inherent visual format; instead, they are edited through rendered *projections*. A single node branch can be viewed as an outline (nested bulleted lists), prose (Markdown headings and paragraphs), or even as a table. This eliminates export friction, as users can seamlessly switch to the most appropriate projection mode to then directly copy and share information.
(2) **Multiple Parents**. Human knowledge is associative and rarely fits into a strict directory hierarchy [5]. Unlike most outliner apps (e.g., Logseq [6] or Tana [7]), Mangrove nodes can have multiple parents, allowing a single piece of information to simultaneously exist in disparate contexts (e.g., a personal daily journal and a shared project outline) without duplication.
(3) **Properties and Schemas**. Nodes support key-value properties and can enforce structural schemas on their children. Combined with projections, this bridges the gap between freeform notes and structured data. Users can interact with otherwise-qualitative text as queryable, columnar data, which is crucial for integrating personal notes with external collaborative tools or AI agents.

Ultimately, this flexible, text-based graph improves individual sensemaking while boosting the collaborative potential of personal notes, enabling serendipitous and AI-driven discoveries across otherwise disparate work infrastructures [8].

2 Design

All information in Mangrove is stored in nodes, where a node contains *content* (e.g., text or a picture), one or more parent nodes, and possibly properties and children nodes. All nodes are part of one tree and are descendants of a singular *root* node (with no parent).

Nodes primarily contain organized information from users, as shown in the following example graph and its corresponding projections in Figure 1:

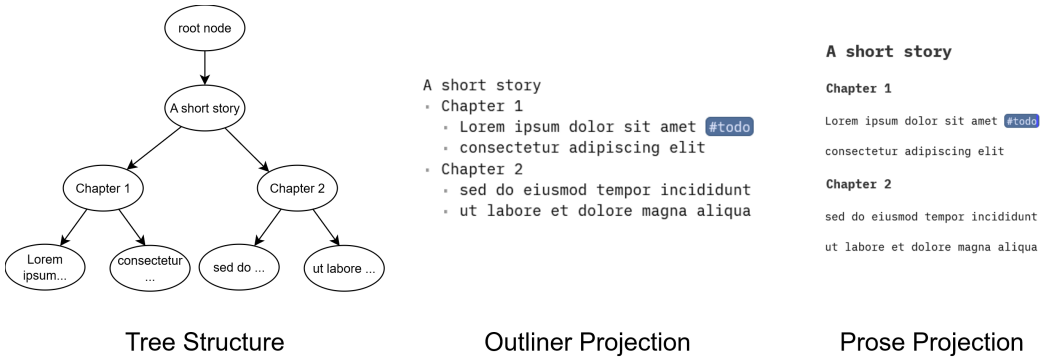


Fig. 1. Example Mangrove tree projected in various ways

Mangrove's editor mimics a standard multi-line text file viewer (supporting common and Vim keybindings), but instead of loading a static file, it *projects* a specific node and its descendants as rendered text (see Figure 1—projections are loaded from the "A short story" node). To decouple format from content, different projection modes dynamically insert formatting characters into temporary node *prefix* and *suffix* buffers (e.g., adding Markdown bullets and spaces for outliner mode, or more newlines for prose). Editor safeguards prevent modification of these buffers, ensuring users only directly edit the underlying content.

Because edits only affect the core content (i.e., editing, adding, or deleting nodes), users can instantly toggle projection modes or 'zoom into' a different node to reproject another branch of the tree. Users can save specific node-and-mode combinations as 'favorite' projections, functioning like frequently visited pages without actually forcing information into rigid folders/pages/files. Mangrove may also generate persistent, synchronized Markdown or CSV files from projections for arbitrary external access. By operating on nodes rather than files, Mangrove eliminates the brittle requirement of links referencing a specific paragraph within a static page to access information (which commonly breaks down).

Unlike traditional outliner apps, Mangrove nodes may have multiple parents (hence the name *Mangrove*, inspired by the coastal tree's distinct aerial roots that appear as if additional branches were extending *below* the trunk). This solves a key problem when trying to reference certain text in multiple locations throughout a PKMS—rather than duplicating content or using links to display non-editable previews of the original content ('transclusions'), a node can simply be included as a child of as many parents as is needed. Any content updates will thus immediately update all active projections of that node.

Multiple parents enable *meta-parents*—special, system-managed nodes that automatically aggregate scattered content. For example, the *Tag* meta-parent automatically groups all nodes containing #todo and other tags through child nodes, while a *Date Created* meta-parent aggregates nodes by timestamp, as depicted in Figure 2. Similar meta-parents could exist for *Date Modified*, *Week Modified*, and more. By projecting these meta-parents, users can instantly generate structured views, such as an *auto-journal* of musings originally scattered across disparate projects across the tree. Beyond aiding human recall, these

automated structural associations help external AI agents parse and recognize connections within the knowledge base.

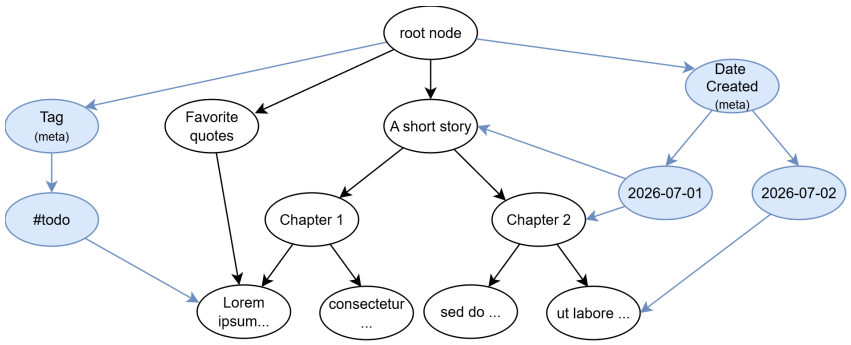


Fig. 2. Multiple parents and meta-parents (blue) in an expanded example Mangrove tree

Nodes also support properties, or key-value pairs that may be optionally projected. Some properties may be automatically managed such as *Date Created* or *Heading Level*, while other properties may be manually set. The *Schema* meta-parent's children each define a particular set of key-value pairs, and nodes with a particular schema as a parent node require their children to conform to the required properties. Schemas can be used for routine data logging, semi-structured data, or even bridge the gap between freeform notes and rigorous research infrastructure. We provide three examples:

(1) A 'book reviews' node with the BookReview schema parent might require its children to include properties such as *author*, *publish date*, and *rating*. Projecting at these node could optionally display said child properties in a tabular or spreadsheet format when convenient or as prose when not, supporting more structured data that can otherwise be hard to cleanly enforce in unfiltered text systems.
(2) A research project branch may contain nodes that claim to answer a research question, and such nodes must have a *research question* property that links to the node defining the question [9].
(3) Automated meta-nodes could mirror PDF highlight/note annotations stored in source citation managers like Zotero. These might use properties following the Web Annotations Data Model [10] that also provide a link to the exact text in the source document. This allows for the creation of rich provenance chains—user notes that reference a "digital twin" source annotation provide a direct traceback to the specific highlighted passage, providing far more context than solely citing the entire paper [9].

As the tree is only editable via projections, the user does not need to choose a note-taking formatting style, as they can switch between prose and outliner modes at will and even display multiple modes at once (e.g., a table rendered in part of an outliner view). Additional projection modes could also be added depending on the information, such as an interactive graph view or a spreadsheet view for nodes with certain property data. There is thus no need for a special 'export' and reformat step when sharing data to other platforms, as the whole projection can be simply copied and pasted in the necessary format.

Neither does the user need to choose a single ontology categorization for their notes—they are free to create associations between nodes as they please and focus on the content.

3 Related Work

Chan et al. [9] detail the current issues in scholarly synthesis, where *ad-hoc* databases of research claims and evidence that are typically disconnected from rigid PDF papers and lose information as a result. Chan's discourse graph solution helps map out the underlying research questions and results using Modular Interoperable Research Attribution (MIRA), and could likely integrate quite well with Mangrove, though Mangrove may also be used more broadly. Various other studies such as Fenlon et al. [3] emphasize the need for and challenge in maintaining a strong knowledge infrastructure between digital communities.

While modern note-taking applications vary widely, two prominent paradigms–outliners and file-based prose–both impose constraints on how knowledge is structured. Outliners like Logseq [6], Tana [7], and Roam Research [11] organize information hierarchically and support some notions of properties and schemas; however, they remain bound to the concept of discrete pages and, crucially, restrict nodes to a single parent. Conversely, prose-focused apps like Obsidian [2] maximize portability by using directories of Markdown files. Yet, this forces information to fit within rigid file boundaries, requiring constant re-indexing to manage cross-file references and relegating properties to per-file YAML frontmatter. While plugins like Dataview/Datacore provide some extensions, they sacrifice portability without fully escaping the underlying file system's limitations. Jung [12] studies PKMS applications and proposes that they can extend human cognition to a second brain (serialized notes) and further into a third brain via AI analysis of said notes, which could also be made more efficient through Mangrove.

4 Implementation & Next Steps

Mangrove is being actively developed as an offline-first Progressive Web App (PWA) to ensure cross-platform accessibility. The app is built on CodeMirror 6 and stored in IndexedDB. Upon reaching maturity, the codebase will be open-sourced.

Mangrove intends to alleviate the burden of formatting, letting users focus squarely on content how it is connected. We hope that these features will reduce technical barriers to better promote collaboration.

A critical next step is robustly ingesting existing notes, such as Markdown directories or pasted HTML. While Mangrove's editor prevents manual edits from breaking projection syntax, importing external data risks introducing malformed structures or syntax. Addressing this will require an automated linter and format decider to automatically convert raw external text into Mangrove's internal node hierarchy.

5 Biography

Ryan West researches distributed information management systems, with a goal of improving individuals' and teams' abilities to understand and innovate on knowledge. He earned his Ph.D. in Computer Science from the University of Utah and remains a part of the Personal Data and Empowerment Lab for accessible Human-Computer Interaction

research. Ryan also operates CodexWest LLC, a boutique, Forbes-featured consultancy offering forensic software expert witness services and other research services.

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